

POTENTIAL OUTCOMES 2035

Aspirations for India's Semiconductor Industry by Niti Aayog

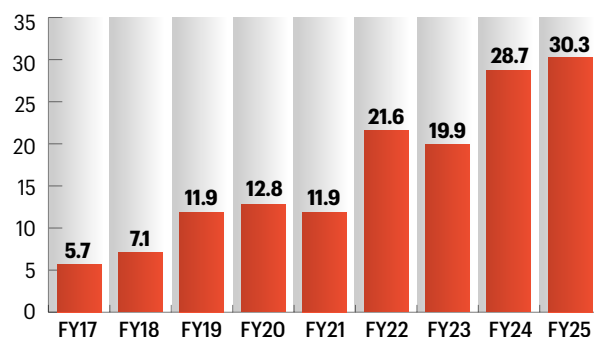
Objective	Target
● Global Market Position	Capture 10-13% of global semiconductor market
● Advanced Packaging (OSAT)	Become a top 3 destination for advanced OSAT capacity
● Compound Semiconductors	Become a leading supplier of SiC and GaN materials
● Chip Self-Sufficiency (2030)	Meet 15–25% of domestic semiconductor demand through local production
● Chip Self-Sufficiency (2035)	Meet 35–50% of domestic semiconductor demand through local production
● Value Self-Sufficiency (2030)	Achieve 35–40% value capture across the semiconductor supply chain
● Value Self-Sufficiency (2035)	Achieve 55–70% value capture across the semiconductor supply chain

SOURCE NITI AAYOG'S FUTURE OF INDIA'S SEMICONDUCTOR INDUSTRY

THE SHOPPING BILL

Semiconductor Imports (\$bn) Up Five Times Since FY17

SOURCE MINISTRY OF COMMERCE AND INDUSTRY



BEYOND THE MACHINES

Building a semiconductor fab is only the beginning. Operating one requires an uninterrupted flow of ultra-pure materials, specialty chemicals and industrial gases that form the backbone of chip manufacturing.

“Etching and deposition gases such as helium, ammonia and fluorine compounds are sourced from Qatar, Algeria, Australia, China and the United States. Specialty gases such as neon, xenon and krypton, critical for semiconductor production, are largely derived as by-products of steelmaking and sourced from China, Ukraine and Russia. Bulk gases such as nitrogen, oxygen, argon and hydrogen are supplied through a global network of industrial gas producers,” says Danish Faruqui, CEO of Fab Economics.

The dependence extends beyond gases. Semiconductor manufacturing also requires wafers, substrates, specialty chemicals, photoresists and packaging materials, many of which are sourced from Taiwan, Japan, Europe, South Korea and the US. Even companies establishing operations in India continue to source critical inputs from overseas.

“To be straightforward, the local ecosystem for semiconductor-grade materials is simply not there yet. Sourcing semiconductor-grade materials locally is indeed one of the most nuanced challenges we face today,” says Ashok Mehta, Founder and Director of Suchi Semicon. The company, setting up an outsourced semiconductor assembly and test (OSAT) facility in Surat, imports equipment from Malaysia, Singapore and Japan.